KS3 DIGITAL TECHNOLOGY

ROBOTICS, DRONES AND SELF-DRIVING VEHICLES

Machines that sense, think and move: how robots work, how drones fly and are regulated, how self-driving vehicles see the road, what robots mean for jobs and safety, and how to program a robot with an algorithm.

LESSONS

1 What is a robot?
2 How robots sense and move
3 Drones
4 Self-driving vehicles
5 Robots and society
6 Program a robot

LESSON 1 OF 6

WHAT IS A ROBOT?

A robot is a programmable machine that can sense its surroundings, make decisions and act in the physical world.

STARTER · 5 MINUTES

DO NOW

Is a washing machine a robot? Is a calculator? Is a self-driving car? Discuss what makes something a robot.

LEARNING INTENTIONS

WE ARE LEARNING TO

Define a robot, describe the sense-think-act cycle and give examples of different types of robot

SUCCESS CRITERIA: I CAN...

- ✓ I can define what a robot is
- ✓ I can describe the sense-think-act cycle
- ✓ I can give examples of different types of robot and where they are used

WHAT MAKES A ROBOT?

- A programmable machine
- Senses its surroundings with sensors
- Makes decisions with a computer (a controller)
- Acts in the physical world with motors and other actuators

KEY WORDS

ROBOT	A programmable machine that can sense its environment, make decisions and act in the physical world.
SENSE-THINK-ACT	The cycle robots follow: gather sensor data, decide what to do, then carry out actions.
COBOT	A collaborative robot designed to work safely beside people.
HUMANOID ROBOT	A robot with a human-like body shape.
AUTOMATION	Using machines to do tasks automatically that people once did by hand.

SENSE, THINK, ACT

- Sense: cameras, distance sensors, touch sensors, microphones
- Think: a program decides what to do next
- Act: wheels turn, arms move, grippers close
- Then sense again: the cycle repeats many times a second

ACTIVITY · 15 MINUTES · PAIRS

ROBOT OR NOT?

1. Sort the 10 picture cards (e.g. robot vacuum, toaster, drone, factory arm, remote-control car, smart speaker) into "robot", "not a robot" and "not sure".
2. For each robot, identify what it senses, how it thinks and how it acts.
3. Agree a one-sentence class definition of a robot.

In your workbook, answer:

? Explain why a robot vacuum is a robot but a toaster isn't.

? Is a remote-control car a robot? Explain.

Your workbook: Lesson 1 · tick each step off as you go

ROBOTS AT WORK

- Industrial arms: welding and painting cars
- Warehouse robots: moving shelves and packing orders
- Surgical robots: helping surgeons make tiny, precise movements
- Cobots work safely beside people; humanoid robots are being trialled in factories

QUICK CHECK

1. What is a robot?

A A website
B A computer game
C Any machine with a plug
D A programmable machine that senses, decides and acts

2. What are the three steps of the robot cycle?

A Walk, run, jump
B Start, stop, pause
C Input, save, print
D Sense, think, act

3. What is a cobot?

A A robot toy
B A robot designed to work safely beside people
C A cloud robot
D A robot cow

4. Which is an example of a surgical robot's advantage?

A It's cheaper than a plaster
B Very small, steady, precise movements
C It never needs a surgeon
D It works without electricity

QUICK CHECK: ANSWERS

1. D: A programmable machine that senses, decides and acts

Robots sense, think and act.

2. D: Sense, think, act

Sense-think-act.

3. B: A robot designed to work safely beside people

Collaborative robot.

4. B: Very small, steady, precise movements

Precision helps keyhole surgery.

2 MARKS · IN YOUR WORKBOOK

EXIT TICKET

Define the term robot.

Look back at today's success criteria: can you tick them all?

- ✓ I can define what a robot is
- ✓ I can describe the sense-think-act cycle
- ✓ I can give examples of different types of robot and where they are used

LESSON 2 OF 6

HOW ROBOTS SENSE AND MOVE

Robots use sensors such as cameras, lidar, ultrasonic sensors, touch sensors and encoders to understand their surroundings and their own position.

STARTER · 5 MINUTES

DO NOW

Close your eyes and touch your nose. How did you know where your hand was?

TODAY

LEARNING INTENTIONS

WE ARE LEARNING TO

Describe the sensors robots use, how they move, and the meaning of degrees of freedom and accuracy

SUCCESS CRITERIA: I CAN...

- ✓ I can name sensors robots use and what each is for
- ✓ I can explain degrees of freedom
- ✓ I can explain why precision and repeatability matter

ROBOT SENSES

- Cameras plus computer vision: recognise objects and people
- Lidar: laser pulses measure distances to build a 3D map
- Ultrasonic and infrared: measure distance to nearby objects
- Touch and force sensors: feel contact and grip strength

KEY WORDS

LIDAR	A sensor that measures distance by timing reflected laser pulses, building a 3D map.
ENCODER	A sensor that counts how far a wheel or joint has turned.
DEGREES OF FREEDOM	The number of independent ways a robot or robot arm can move.
END EFFECTOR	The tool on the end of a robot arm, such as a gripper or welder.
COMPUTER VISION	AI that lets computers understand images and video.

KNOWING WHERE YOU ARE

- Encoders count how far each wheel or joint has turned
- IMUs (accelerometers and gyroscopes) sense tilt and movement
- GPS for outdoor robots; maps for indoor ones
- Combining sensors gives a more reliable picture

MOVING

- Motors turn wheels, joints and propellers
- Degrees of freedom: the number of independent ways a robot can move
- A typical factory arm has 6 degrees of freedom
- The end effector is the "hand": a gripper, welder, suction cup or tool

ACTIVITY · 20 MINUTES · PAIRS

SENSOR MATCH-UP

1. Read the six robot missions (e.g. a robot that picks ripe strawberries, a robot that cleans a swimming pool, a delivery robot crossing a pavement).
2. Choose the sensors each robot needs and explain why.
3. Choose an end effector for each robot that has an arm.
4. Identify one situation where a sensor could be fooled, and which other sensor would help.

In your workbook, answer:

? Which sensors does a strawberry-picking robot need, and why?

? Describe a situation where a sensor could be fooled.

Your workbook: Lesson 2 · tick each step off as you go

QUICK CHECK

1. What does lidar use to measure distance?

A Smell
B Magnets
C Laser pulses
D Sound only

2. What does an encoder measure?

A How far a wheel or joint has turned
B Colour
C Temperature
D Sound volume

3. What are degrees of freedom?

A The number of independent ways a robot can move
B How free the robot is to leave
C The robot's price
D Its speed

4. What is an end effector?

A The tool on the end of a robot arm
B The robot's power switch
C A type of sensor
D The final line of code

QUICK CHECK: ANSWERS

1. **C: Laser pulses**
Lidar times laser reflections.

2. **A: How far a wheel or joint has turned**
Encoders count rotation.

3. **A: The number of independent ways a robot can move**
Typical arms have 6.

4. **A: The tool on the end of a robot arm**
Grippers, welders, suction cups.

2 MARKS · IN YOUR WORKBOOK

EXIT TICKET

Explain why a robot might use more than one type of sensor.

Look back at today's success criteria: can you tick them all?

- ✓ I can name sensors robots use and what each is for
- ✓ I can explain degrees of freedom
- ✓ I can explain why precision and repeatability matter

LESSON 3 OF 6

DRONES

Most drones are quadcopters that steer by changing the speed of four rotors.

STARTER · 5 MINUTES

DO NOW

Name three jobs a drone could do better, faster or more safely than a person.

TODAY

LEARNING INTENTIONS

WE ARE LEARNING TO

Explain how drones fly, describe their uses and apply the UK rules for flying a drone safely

SUCCESS CRITERIA: I CAN...

- ✓ I can explain how a quadcopter moves
- ✓ I can describe uses of drones
- ✓ I can apply the UK Drone Code

HOW A QUADCOPTER FLIES

- Four rotors: two spin clockwise, two anticlockwise
- All faster: rise; all slower: descend
- Tilt by speeding up rotors on one side: the drone moves that way
- A flight controller with an IMU makes hundreds of tiny adjustments a second

KEY WORDS

DRONE	An aircraft with no pilot on board, controlled remotely or flying itself.
QUADCOPTER	A drone with four rotors.
FLYER ID	Proof from the UK Civil Aviation Authority that someone has passed the online drone test.
OPERATOR ID	A UK registration for the person responsible for a drone, which must be displayed on it.
VISUAL LINE OF SIGHT	Being able to see your drone with your own eyes at all times.

DRONES AT WORK

- Search and rescue: thermal cameras find missing people
- Farming: crop monitoring and precision spraying
- Inspections: wind turbines, bridges, power lines
- Deliveries: medicines and blood samples between hospitals; trials of parcel delivery

THE DRONE CODE (UK)

- Get your Flyer ID (free online test) and Operator ID from the CAA
- Never fly higher than 120 m (400 ft)
- Keep the drone in sight at all times
- Stay away from airports, crowds and people; respect privacy

ACTIVITY · 25 MINUTES · GROUPS OF 3

MISSION CONTROL

1. Your team is planning a drone mission to find a lost walker on Cuilcagh Mountain.
2. Choose the drone features you need (camera type, battery, GPS, speaker).
3. Plan your flight using the Drone Code: height, line of sight, people and permissions.
4. Identify two risks and how you'd reduce them.

In your workbook, answer:

? Which drone features do you need, and why?

? Describe two risks and how you'd reduce them.

Your workbook: Lesson 3 · tick each step off as you go

QUICK CHECK

1. How many rotors does a quadcopter have?

A Two
B One
C Four
D Six

2. What is the maximum height for most drone flights in the UK?

A 10 metres
B 1,000 metres
C 120 metres (400 ft)
D No limit

3. What does visual line of sight mean?

A The drone must face you
B You can fly it anywhere
C You must film in straight lines
D You must be able to see the drone at all times

4. Which organisation sets UK drone rules?

A The BBC
B Ofcom
C The NHS
D The Civil Aviation Authority (CAA)

QUICK CHECK: ANSWERS

1. C: Four

Quad = four.

2. C: 120 metres (400 ft)

Stay below 120 m.

3. D: You must be able to see the drone at all times

Keep it in sight.

4. D: The Civil Aviation Authority (CAA)

CAA regulates flying.

2 MARKS · IN YOUR WORKBOOK

EXIT TICKET

Explain how a quadcopter rises and moves forward.

Look back at today's success criteria: can you tick them all?

- ✓ I can explain how a quadcopter moves
- ✓ I can describe uses of drones
- ✓ I can apply the UK Drone Code

LESSON 4 OF 6

SELF-DRIVING VEHICLES

Self-driving vehicles use cameras, radar, lidar and AI to perceive the road, predict what others will do, plan a path and control the car.

STARTER · 5 MINUTES

DO NOW

What does a human driver need to notice in the first ten seconds of pulling out of a junction?

TODAY

LEARNING INTENTIONS

WE ARE LEARNING TO

Describe how self-driving vehicles perceive, plan and act, and explain the levels of driving automation

SUCCESS CRITERIA: I CAN...

- ✓ I can describe how a self-driving car senses its surroundings
- ✓ I can explain the levels of driving automation
- ✓ I can describe the benefits and challenges of self-driving vehicles

HOW A SELF-DRIVING CAR "SEES"

- Cameras: colours, lanes, signs and traffic lights
- Radar: distance and speed of other vehicles, even in rain and fog
- Lidar: a precise 3D map of everything around the car
- AI combines them to perceive, predict, plan and control

KEY WORDS

AUTONOMOUS VEHICLE	A vehicle that can drive itself without human control.
RADAR	A sensor that uses radio waves to detect the distance and speed of objects.
LEVELS OF AUTOMATION	A scale from 0 to 5 describing how much of the driving a vehicle does itself.
EDGE CASE	A rare or unusual situation that a system finds hard to handle.
ROBOTAXI	A taxi that drives itself with no human driver.

LEVELS OF DRIVING AUTOMATION

- Level 0-2: the human drives; the car helps (e.g. lane-keeping, adaptive cruise control)
- Level 3: the car drives in some conditions; the human must take over when asked
- Level 4: the car drives itself fully in a defined area, e.g. a robotaxi zone
- Level 5: drives anywhere, any conditions (not yet achieved)

THE ROAD AHEAD

- Robotaxis already carry paying passengers in some US and Chinese cities
- UK: the Automated Vehicles Act 2024; pilots of self-driving taxi services
- When the car drives itself, the company, not the passenger, is responsible
- Challenges: edge cases, bad weather, cost, trust, jobs

ACTIVITY · 20 MINUTES · PAIRS

EDGE CASE CHALLENGE

1. Read the eight driving scenarios (e.g. a ball rolling into the road, a tractor on a narrow country lane, roadworks with a person holding a stop sign).
2. For each, decide what the car must perceive, predict and do.
3. Rate how hard each would be for a self-driving car (1-5) and explain why.
4. Decide which would be hardest on Fermanagh's rural roads.

In your workbook, answer:

? What should the car predict when a ball rolls into the road?

? Which scenario is hardest on rural roads, and why?

Your workbook: Lesson 4 · tick each step off as you go

QUICK CHECK

1. Which sensor works best in fog and rain?

A Camera
B Lidar
C A microphone
D Radar

2. At which level does the car drive fully, but only in a defined area?

A Level 1
B Level 2
C Level 0
D Level 4

3. What is an edge case?

A A rare, unusual situation that's hard to handle
B A type of sensor
C A phone case
D The edge of the road

4. Under the UK Automated Vehicles Act 2024, who is responsible when the vehicle is driving itself?

A The other drivers
B The authorised company
C The passenger
D Nobody

QUICK CHECK: ANSWERS

1. **D: Radar**
Radio waves pass through fog.

2. **D: Level 4**
Level 4 robotaxis are geofenced.

3. **A: A rare, unusual situation that's hard to handle**
Edge cases challenge AI.

4. **B: The authorised company**
Responsibility moves to the company.

4 MARKS · IN YOUR WORKBOOK

EXIT TICKET

Describe the four stages a self-driving car's software goes through.

Look back at today's success criteria: can you tick them all?

- ✓ I can describe how a self-driving car senses its surroundings
- ✓ I can explain the levels of driving automation
- ✓ I can describe the benefits and challenges of self-driving vehicles

LESSON 5 OF 6

ROBOTS AND SOCIETY

Robots can take on dull, dirty and dangerous work and boost productivity, but automation also changes and replaces jobs.

STARTER · 5 MINUTES

DO NOW

Would you rather a robot or a person did these jobs: bin collector, nurse, surgeon, teacher, soldier? Why?

TODAY

LEARNING INTENTIONS

WE ARE LEARNING TO

Evaluate the impact of robots and automation on jobs, safety and society

SUCCESS CRITERIA: I CAN...

- ✓ I can describe the impact of robots on jobs
- ✓ I can discuss safety and responsibility when robots go wrong
- ✓ I can form a reasoned opinion on an ethical question about robots

THE THREE DS

- Robots are ideal for dull, dirty and dangerous work
- Dull: repetitive packing and sorting
- Dirty: sewer inspection, cleaning tanks
- Dangerous: bomb disposal, nuclear clean-up, deep sea, space

KEY WORDS

DULL, DIRTY AND DANGEROUS	The kinds of work robots are especially suited to.
JOB DISPLACEMENT	When people's jobs are replaced by machines or automation.
ACCOUNTABILITY	Being responsible for what happens and answering for it.
ETHICS	Deciding what is morally right and wrong.

ROBOTS AND JOBS

- Some jobs disappear or shrink (job displacement)
- Many jobs change: people work alongside robots
- New jobs appear: robot technicians, programmers, designers
- The big question: who benefits, and who gets retrained?

WHO'S RESPONSIBLE?

- If a robot causes harm, who is accountable: maker, programmer, owner or user?
- Safety standards, testing and emergency stops
- Some decisions may need a "human in the loop"
- Should robots ever make life-or-death decisions?

ACTIVITY · 30 MINUTES · GROUPS OF 4

ROBOT DEBATE

1. Your group is given a motion, e.g. "Care homes should use robots to look after residents" or "Delivery robots should be allowed on all pavements".
2. Prepare two arguments for and two against, using evidence from this topic.
3. Hold a short debate with another group.
4. Write your own conclusion, giving reasons.

In your workbook, answer:

? Write your strongest argument for and against your motion.

? Write your own conclusion.

Your workbook: Lesson 5 · tick each step off as you go

QUICK CHECK

1. What are the "three Ds" of robot work?

A Dull, dirty and dangerous
B Drive, drop and dig
C Dark, damp and deep
D Digital, data and design

2. What is job displacement?

A Getting a new job title
B Working from home
C People's jobs being replaced by machines
D Moving desks around

3. Which job has been created by robotics?

A Switchboard operator
B Robot technician
C Lamplighter
D Town crier

4. What does accountability mean?

A Keeping a diary
B Having an account
C Being responsible for what happens
D Counting money

QUICK CHECK: ANSWERS

1. **A: Dull, dirty and dangerous**
   Ideal tasks for robots.

2. **C: People's jobs being replaced by machines**
   Automation can replace jobs.

3. **B: Robot technician**
   New tech creates new jobs.

4. **C: Being responsible for what happens**
   Who is to blame if harm occurs?

2 MARKS · IN YOUR WORKBOOK

EXIT TICKET

Give two examples of dangerous jobs that robots can do.

Look back at today's success criteria: can you tick them all?

- ✓ I can describe the impact of robots on jobs
- ✓ I can discuss safety and responsibility when robots go wrong
- ✓ I can form a reasoned opinion on an ethical question about robots

LESSON 6 OF 6

PROGRAM A ROBOT

Robots follow algorithms.

ROBOTICS, DRONES AND SELF-DRIVING VEHICLES

STARTER · 5 MINUTES

DO NOW

Write instructions to guide a blindfolded partner from the door to your desk. What went wrong?

TODAY

LEARNING INTENTIONS

WE ARE LEARNING TO

Design, write and test an algorithm to control a robot using sequence, selection and iteration

SUCCESS CRITERIA: I CAN...

- ✓ I can write an algorithm to control a robot
- ✓ I can use sequence, selection and iteration with sensor inputs
- ✓ I can test and debug a robot program

ROBOTS FOLLOW ALGORITHMS

- An algorithm is a precise set of steps
- Sequence: steps in order (forward 50 cm, turn right 90°)
- Selection: IF a sensor detects something THEN do this ELSE do that
- Iteration: repeat (a loop) so the robot keeps reacting

KEY WORDS

ALGORITHM	A precise set of steps to solve a problem.
SELECTION	Making a decision in a program using IF ... THEN ... ELSE.
ITERATION	Repeating instructions in a program using a loop.
DEBUGGING	Finding and fixing errors in a program.
SIMULATION	A computer model used to test something safely before doing it for real.

LINE-FOLLOWER

- Two light sensors look down at a black line
- IF left sensor sees black THEN turn left
- ELSE IF right sensor sees black THEN turn right
- ELSE go forward ... REPEAT forever

TEST AND DEBUG

- Test in a simulation first
- Then test on the real robot, one part at a time
- Record what you expected vs what happened
- Real robots slip, batteries fade, and lighting changes sensor readings

ACTIVITY · 35 MINUTES · PAIRS

CODE THE ROBOT

1. Choose a challenge: escape a maze, follow a line, or patrol a room avoiding obstacles.

2. Write your algorithm in pseudocode or a flowchart, using sequence, selection and iteration.

3. Test it (in a simulator, with a micro:bit buggy, or by "playing robot" with a partner).

4. Record one bug, how you found it and how you fixed it.

In your workbook, answer:

? Write your algorithm.

? Describe one bug and how you fixed it.

Your workbook: Lesson 6 · tick each step off as you go

QUICK CHECK

1. What does selection mean in a robot program?

A Making a decision with IF ... THEN ... ELSE
B Running steps in order
C Repeating steps
D Choosing a robot

2. Why do robot programs usually use a loop?

A To make the code longer
B So the robot keeps sensing and reacting
C To save battery
D To stop the robot

3. How does a line-following robot detect a black line?

A It listens for the line
B Light sensors detect less reflected light
C GPS
D A camera reads "line"

4. Why test a robot in a simulation first?

A Simulations are always perfect
B It avoids writing code
C It's quicker, cheaper and safer
D Real robots don't need testing

QUICK CHECK: ANSWERS

1. **A: Making a decision with IF … THEN … ELSE**
   Decisions use sensor data.

2. **B: So the robot keeps sensing and reacting**
   Iteration keeps it responsive.

3. **B: Light sensors detect less reflected light**
   Black reflects less light.

4. **C: It's quicker, cheaper and safer**
   Then test for real.

3 MARKS · IN YOUR WORKBOOK

EXIT TICKET

Write an algorithm for a robot that stops when an object is less than 10 cm away and otherwise moves forward.

Look back at today's success criteria: can you tick them all?

- ✓ I can write an algorithm to control a robot
- ✓ I can use sequence, selection and iteration with sensor inputs
- ✓ I can test and debug a robot program

END OF TOPIC

ROBOTICS, DRONES AND SELF-DRIVING VEHICLES

Revise each lesson in your workbook: the notes, key words and quick checks.
Then take the topic test when Mr Butler sets it.